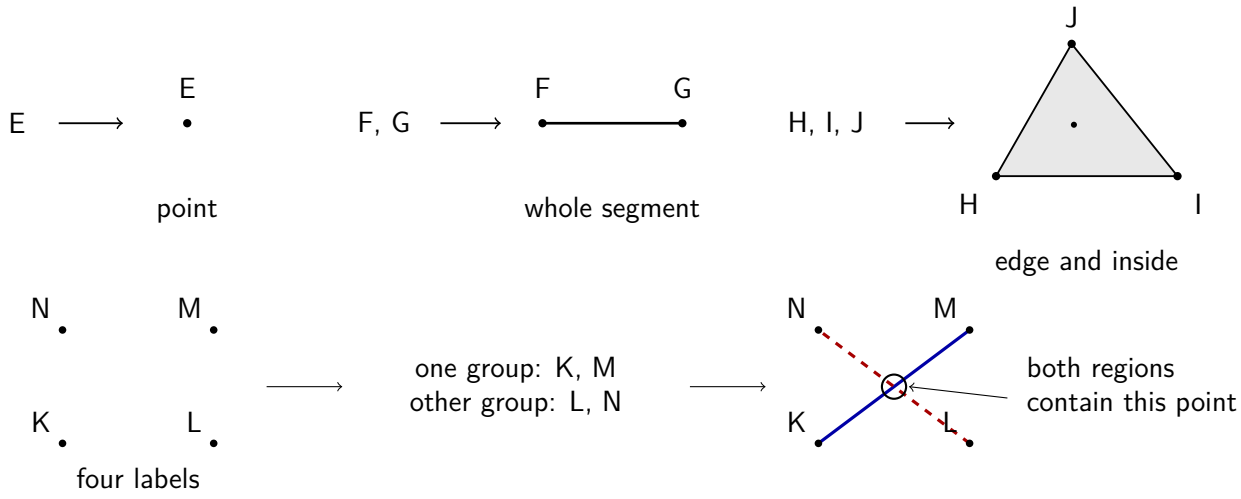
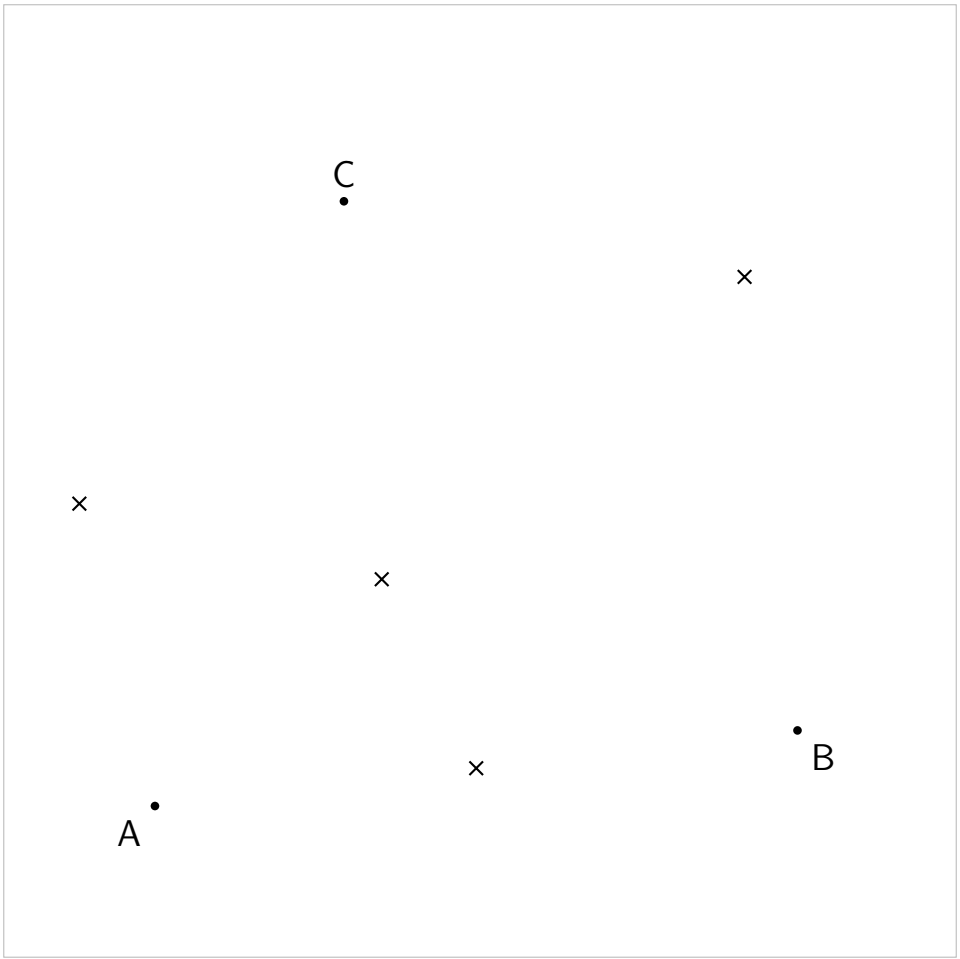
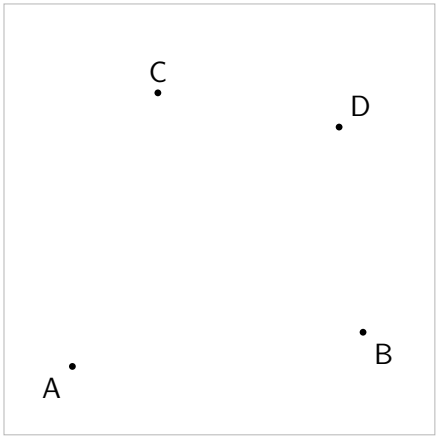
Use each label once in two groups, with at least one label in each. Draw each group's stretched-band region and lightly shade its inside. Use the centers of the marks. Sharing even one point counts. Swapping the two groups is the same split.

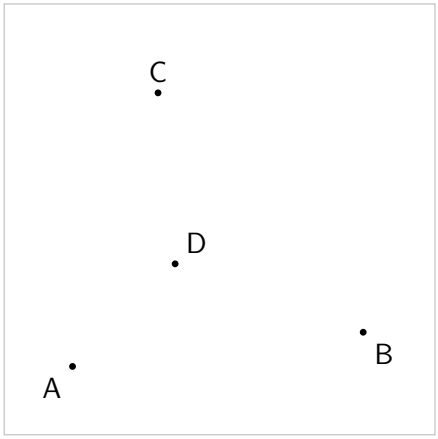


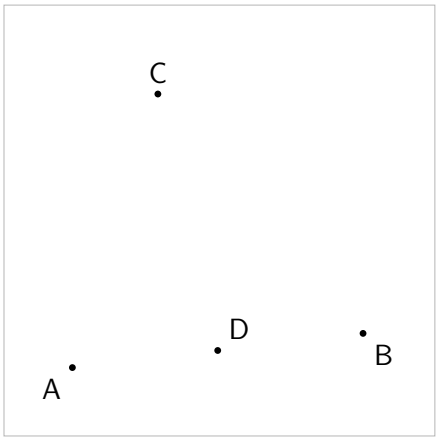
Problem 1: Put D on each cross. For each position, find a split of A, B, C, and D whose two regions meet.

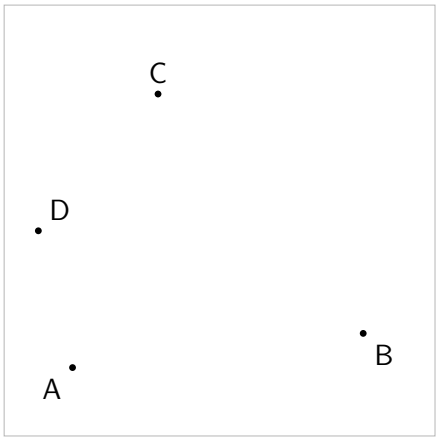


Problem 1: For each position of D, draw the two regions of your split and mark a shared point.

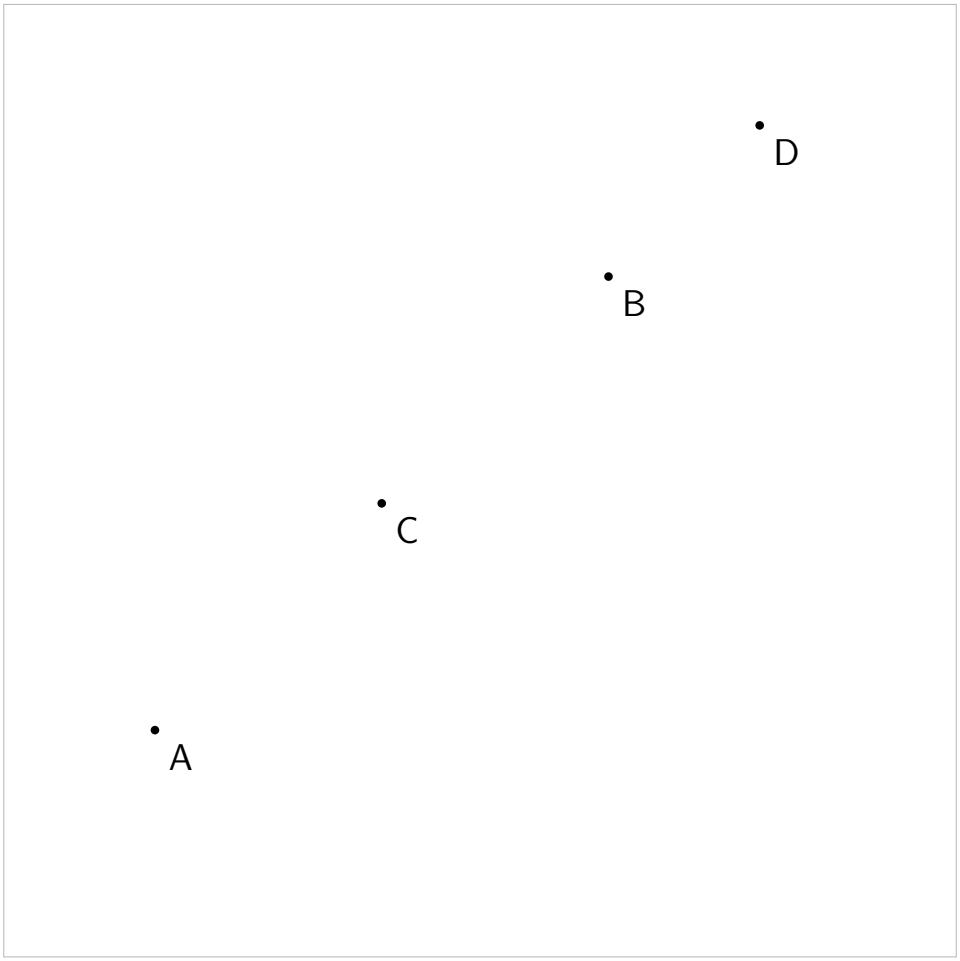




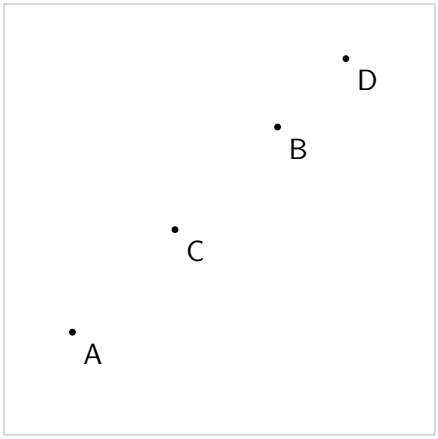


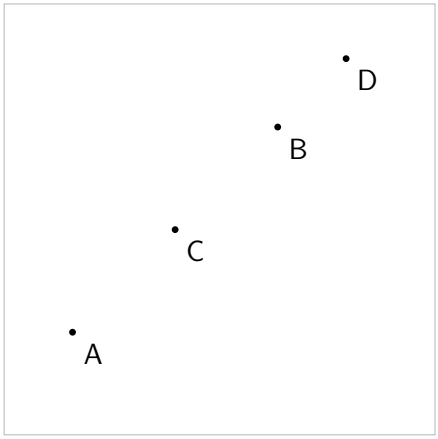


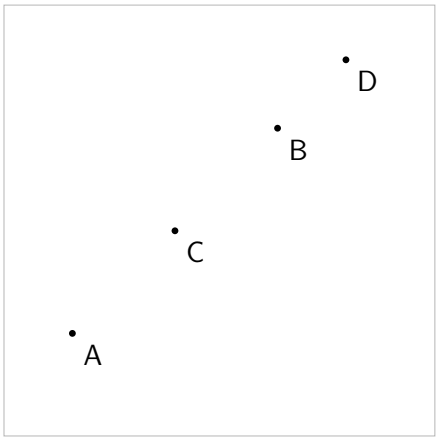
Problem 2: Find every split whose regions meet. Explain how you know your list is complete.

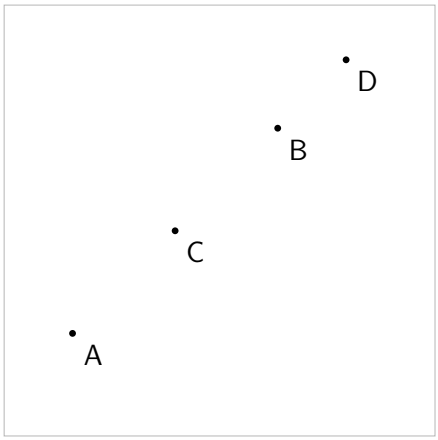


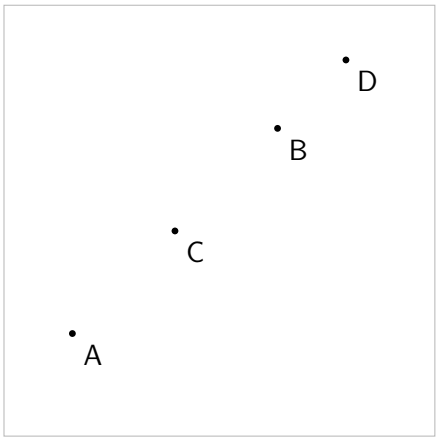
Problem 2: Find every split whose regions meet. Explain how you know your list is complete.

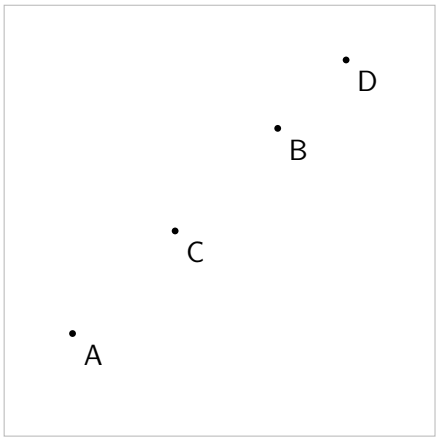












Problem 3: Take turns placing A, B, C, and D in the box. Try to make an arrangement with no successful split. Decide whether your opponent must always be able to find one.